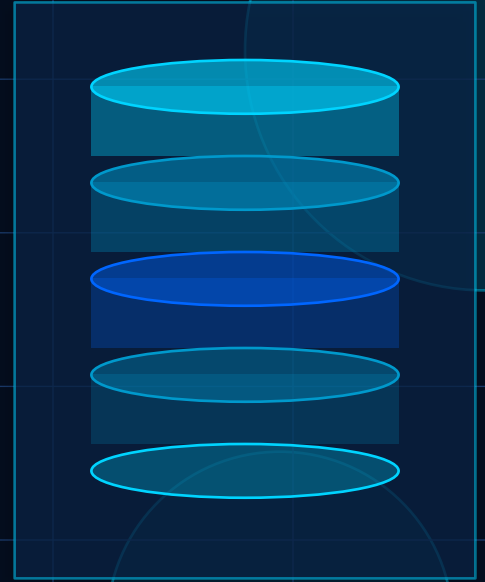
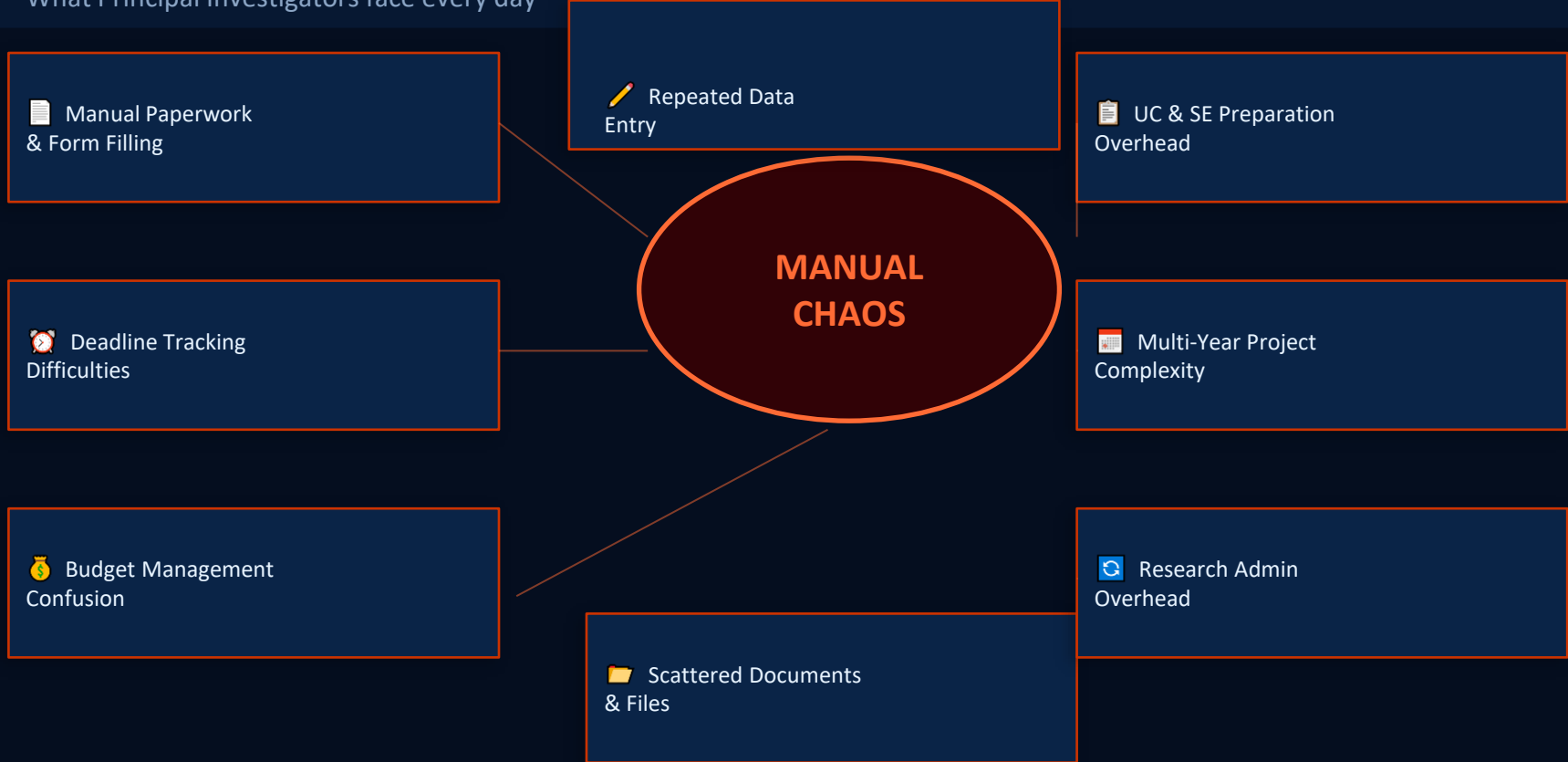

AI-Powered Research Grant Management System

Backend Architecture, Database Engineering & Automated Research Workflow System



THE REAL PROBLEM

What Principal Investigators face every day



WHAT IS ANRF?

Anusandhan National Research Foundation — India's premier research funding body

ANRF Overview

- Apex body for R&D funding in India
- Funds science, tech & innovation
- Successor to SERB framework
- Connects industry & academia
- Annual & multi-year project grants

ANRF Funding Lifecycle



⚡ ANRF requires detailed financial records, UC certificates, and annual progress reports for every funded project

PROJECT OBJECTIVE

"To create an centralized backend system that automates research project management, financial tracking, document handling, and ANRF form generation for Principal Investigators."



Automation

End-to-end workflow automation



AI Assistance

Smart summaries & suggestions



Backend Intel.

Database-driven workflow engine



Real-world Ready

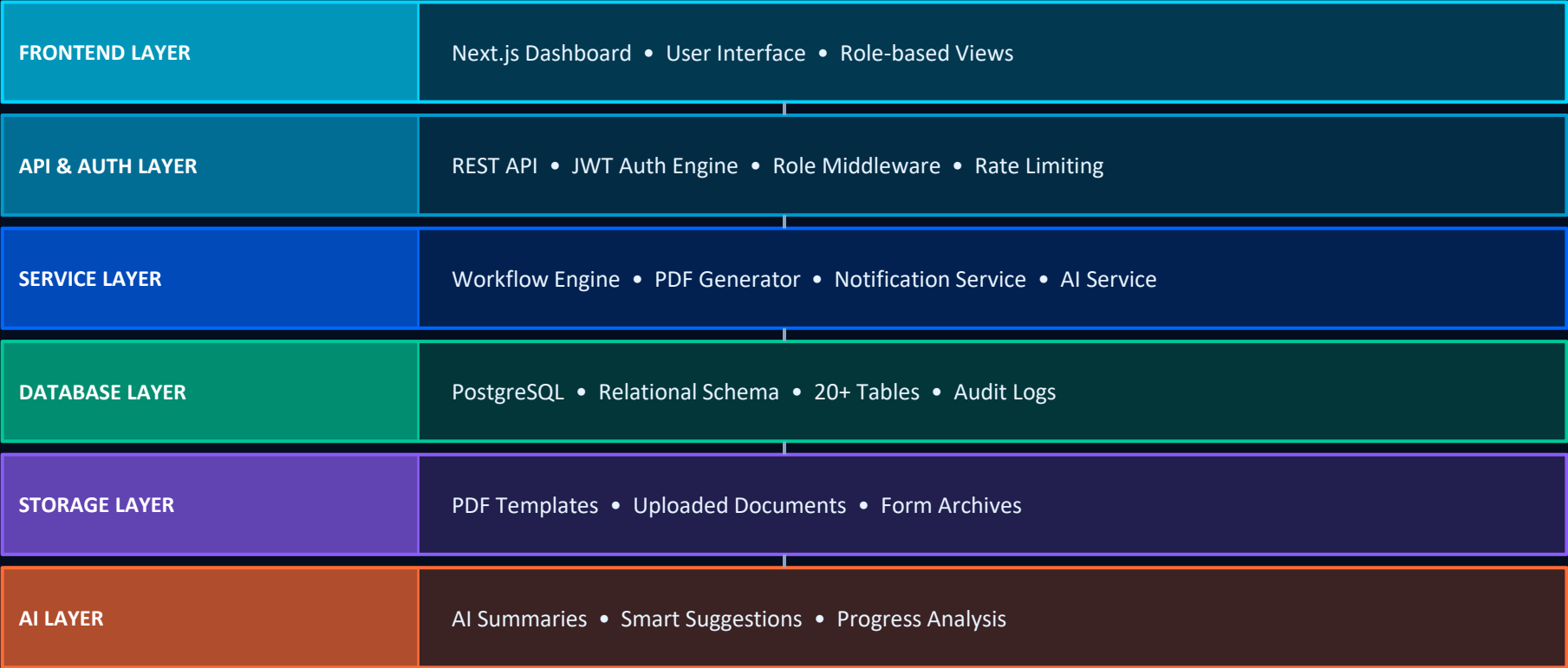
ANRF-aligned deployment



PDF Engine

Auto-form filling system

SYSTEM ARCHITECTURE OVERVIEW



USER ROLE HIERARCHY

PI — Principal Investigator

- Project Owner & Authority
- Creates & manages project
- Generates ANRF forms
- Manages team & finances
- Tracks deadlines & reports
- Approves all documents

Co-PI — Collaborator

- Project collaborator
- Uploads progress data
- Views project status
- Access to reports
- Shared visibility

JRF / SRF — Research Staff

- Research assistant role
- Uploads work & logs
- Activity tracking
- Document submission
- Limited project view

⚠ No Separate Admin — The PI is the Project Authority. Role-based database access ensures data integrity.

6-LAYER DATABASE ARCHITECTURE

1 Identity & Access Layer

users • institutions • roles • permissions

2 Project Core Layer

projects • project_members • funding_agencies • scheme_types

3 Financial System Layer

budget_heads • fund_releases • expenses • assets • manpower

4 Document Management Layer

documents • form_templates • generated_forms • submissions

5 Reporting & Notifications Layer

progress_reports • publications • deadlines • notifications

6 AI & Audit Layer

ai_interactions • audit_logs • form_field_mappings

CORE DATABASE TABLES — 20 Tables, Zero Orphan Data

Identity users	Identity institutions	Identity funding_agencies	Core projects
Core project_members	Finance budget_heads	Finance fund_releases	Finance expenses
Finance manpower_payments	Finance assets	Docs documents	Docs form_templates
Docs form_field_mappings	Docs generated_forms	Reports publications	Reports progress_reports
Reports deadlines	Reports notifications	AI ai_interactions	AI audit_logs

USERS TABLE — Identity & ANRF Profile



TABLE: users

user_id	UUID PK
full_name	VARCHAR
email	VARCHAR UNIQUE
designation	VARCHAR
pan_number	VARCHAR
orcid_id	VARCHAR
highest_degree	VARCHAR
institution_id	UUID FK
password_hash	TEXT
role	ENUM

Why These Fields Exist

PAN Number	Required for ANRF financial disbursement
ORCID ID	Researcher identity in publication databases
Designation	Appears on all official ANRF forms
Highest Degree	PI eligibility verification for grants
Institution FK	Multi-institution deployment support
Role ENUM	Controls access at DB level
Password Hash	bcrypt secured, never plain-text

PROJECTS TABLE — Central Hub of the System

Schema Fields

project_id	UUID PK
title	VARCHAR
sanction_number	VARCHAR
funding_agency_id	FK
scheme_type	ENUM
sanctioned_amount	DECIMAL
recurring_amount	DECIMAL
non_recurring_amount	DECIMAL
start_date	DATE
end_date	DATE
extended_end_date	DATE
project_status	ENUM



expenses

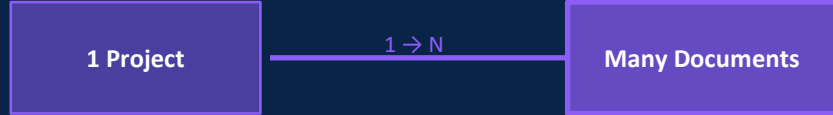
documents

deadlines

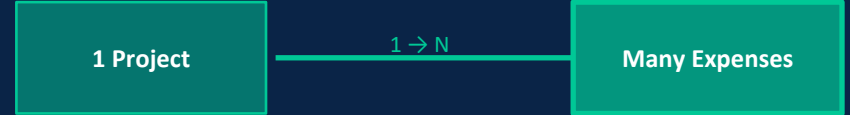
fund_releases

progress_reports

RELATIONSHIP LOGIC — ER Design Principles



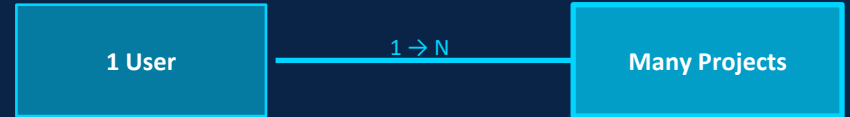
All project files linked by project_id FK



Financial trail attached to each project



Auto-generated deadline schedule per project



PI can manage multiple research projects

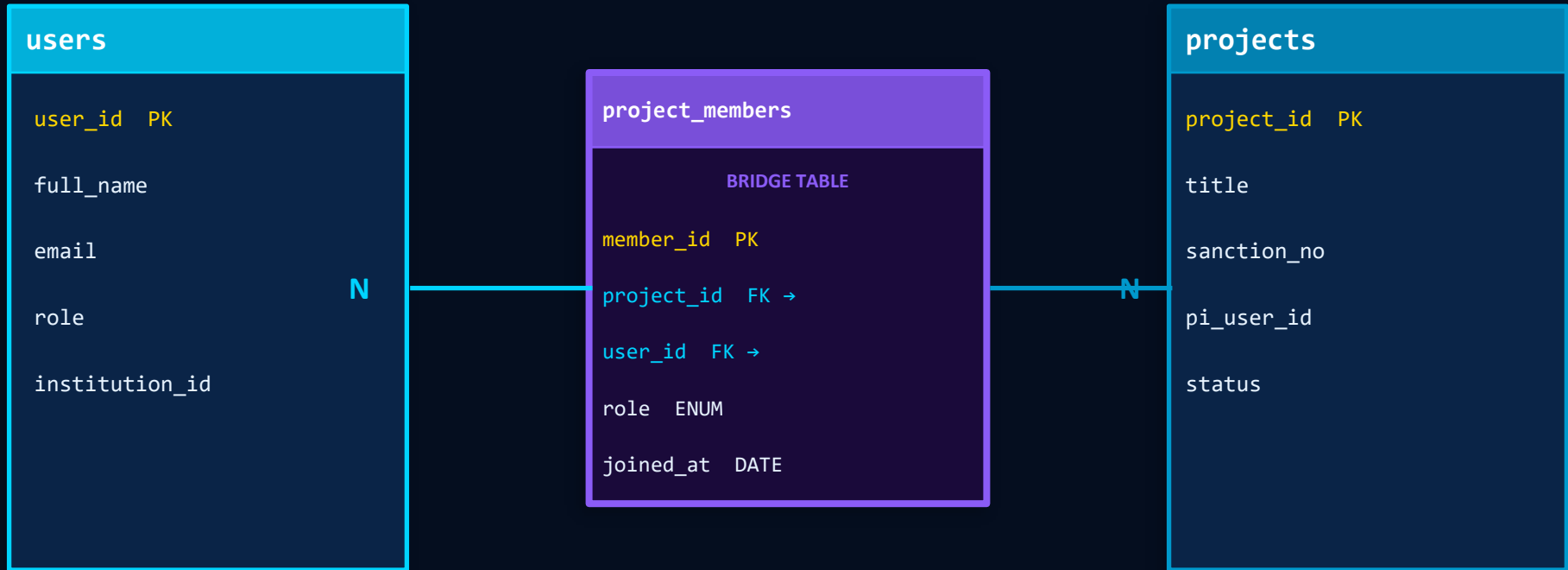


Team structure via bridge table



Annual progress reports tracked per year

BRIDGE TABLE CONCEPT — Many-to-Many Resolution



One PI → Many Projects • One Project → Many Researchers • Bridge table resolves M:N relationship cleanly

FINANCIAL MANAGEMENT SYSTEM

budget_heads

Recurring vs Non-Recurring allocation per head.
Prevents cross-head overspending.

Manpower • Equipment • Travel • Consumables •
Overhead

fund_releases

Annual fund disbursement tracking.
Installment-wise fund arrival records.

Installment 1 • Installment 2 • Year tracking • Release
date • Amount

expenses

Every payment against a budget head.
Linked to bills, vouchers, documents.

vendor_name • amount • budget_head_fk •
voucher_no • date

assets

Non-recurring equipment & asset tracking.
Required for ANRF asset statement.

asset_name • purchase_date • cost • location • condition

manpower_payments

JRF/SRF/RA salary disbursements.
Separate from general expenses.

person_name • month • gross_pay • deductions • net_pay



Every expense is linked to a budget head → project → PI. Full financial traceability for government audit.

AUTOMATED DEADLINE ENGINE

UC Deadlines

Utilization Certificate

- Year 1 UC → 6 months after fund release
- Year 2 UC → Same pattern
- Final UC → 3 months after project end

APR Deadlines

Annual Progress Report

- APR Year 1 → 12 months after start
- APR Year 2 → 24 months after start
- Continuation until project end

PCR Deadline

Project Completion Report

- PCR → 6 months before project end
- Final submission reminder
- Extension request window



Smart Reminder Engine

30 Days
First notice sent

15 Days
Escalation alert

7 Days
Urgent warning

0 Days
Overdue flag

DOCUMENT MANAGEMENT SYSTEM



Sanction Orders

Original ANRF approval docs



Expenditure Bills

Vendor invoices & vouchers



UC Uploads

Utilization certificates



Progress Reports

Annual research reports



Certificates

Completion & compliance docs



Publications

Research papers & patents

System Features

Version Tracking

Submission Status

File History

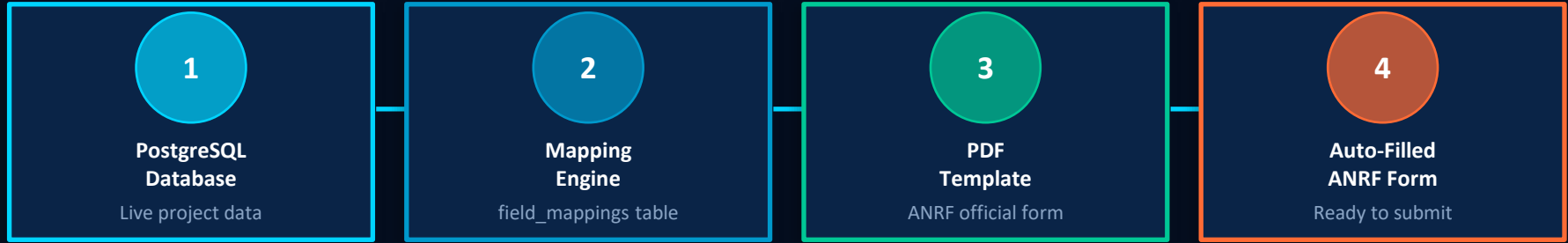
Audit Trail

Download Links

Review Workflow

PDF AUTO-FILL SYSTEM ★ Most Powerful Feature

"Zero Manual Typing — Every ANRF form is auto-populated from the database"



✗ BEFORE: Manual Process

- Print form • Fill by hand • Risk of errors
- Look up data • Recopy from files
- Hours of repetitive data entry
- Missing fields cause rejection

✓ AFTER: Automated System

- Click → Form generated instantly
- DB values mapped automatically
- Zero data entry errors
- Consistent format every time

TYPE 1 — Field-Based PDF Filling

For PDFs with embedded interactive form fields — uses field name matching for precise, fast filling



PDF with Embedded Fields

Form has AcroForm fields with unique field names



Field Name Extraction

System reads all field names from PDF



DB Value Lookup

Match field name to database column



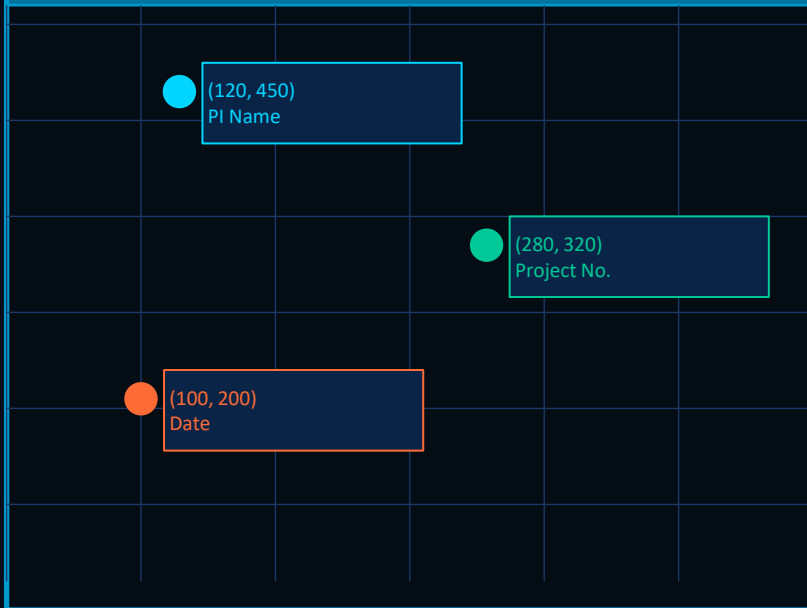
Auto-Fill & Export

Values stamped, PDF ready to sign

TYPE 2 — Coordinate-Based PDF Filling

For plain/scanned government PDFs with no embedded fields — uses x,y coordinate text stamping

PDF Coordinate Grid



How Coordinate Filling Works

- 1 Upload plain/scanned ANRF PDF to system
- 2 Admin maps each field → x,y coordinate (one-time setup)
- 3 Coordinates stored in form_field_mappings table
- 4 When form generated: system reads DB values + coordinates
- 5 Text is stamped at exact pixel positions on PDF
- 6 Final PDF exported — ready for official submission

FORM FIELD MAPPING ENGINE

The form_field_mappings table is the bridge between database values and PDF output

form_field_mappings

mapping_id	UUID PK
template_id	UUID FK
field_name	VARCHAR
db_source	VARCHAR
field_type	ENUM
x_coordinate	FLOAT
y_coordinate	FLOAT
page_number	INTEGER
format_mask	VARCHAR

How It Bridges DB → PDF

projects.title → field: 'project_title'
Project name from DB

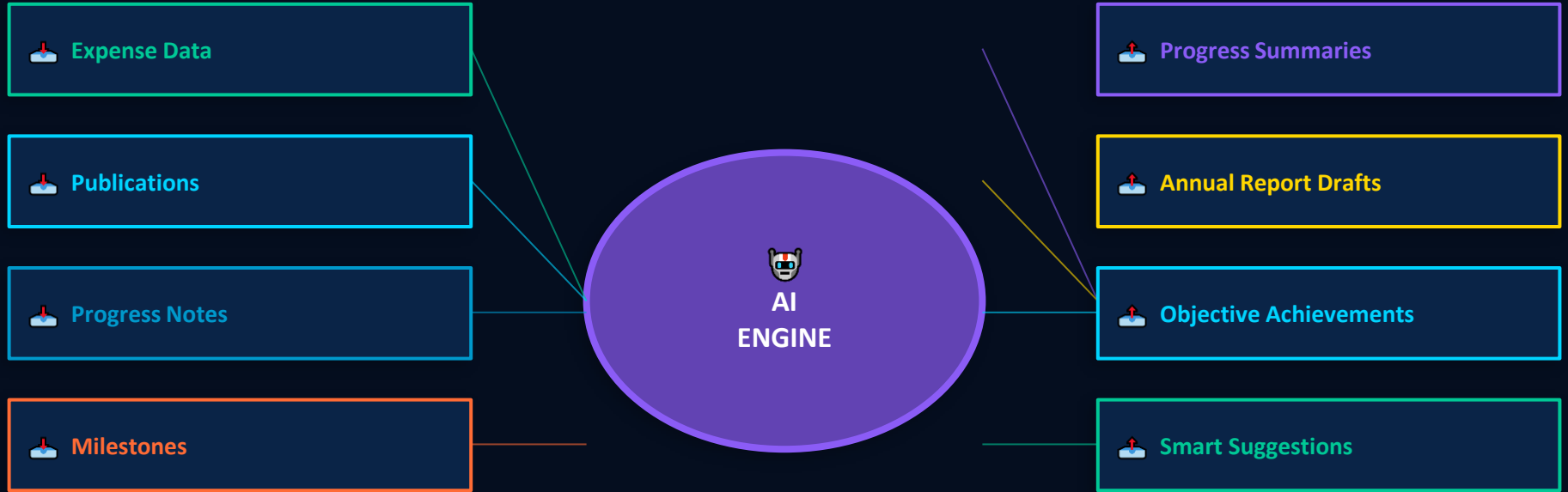
users.pan_number → field: 'pi_pan'
PI PAN for tax forms

fund_releases.amount → coord: (245, 380)
Fund amount at position

expenses.total → field: 'total_exp'
Total expenditure

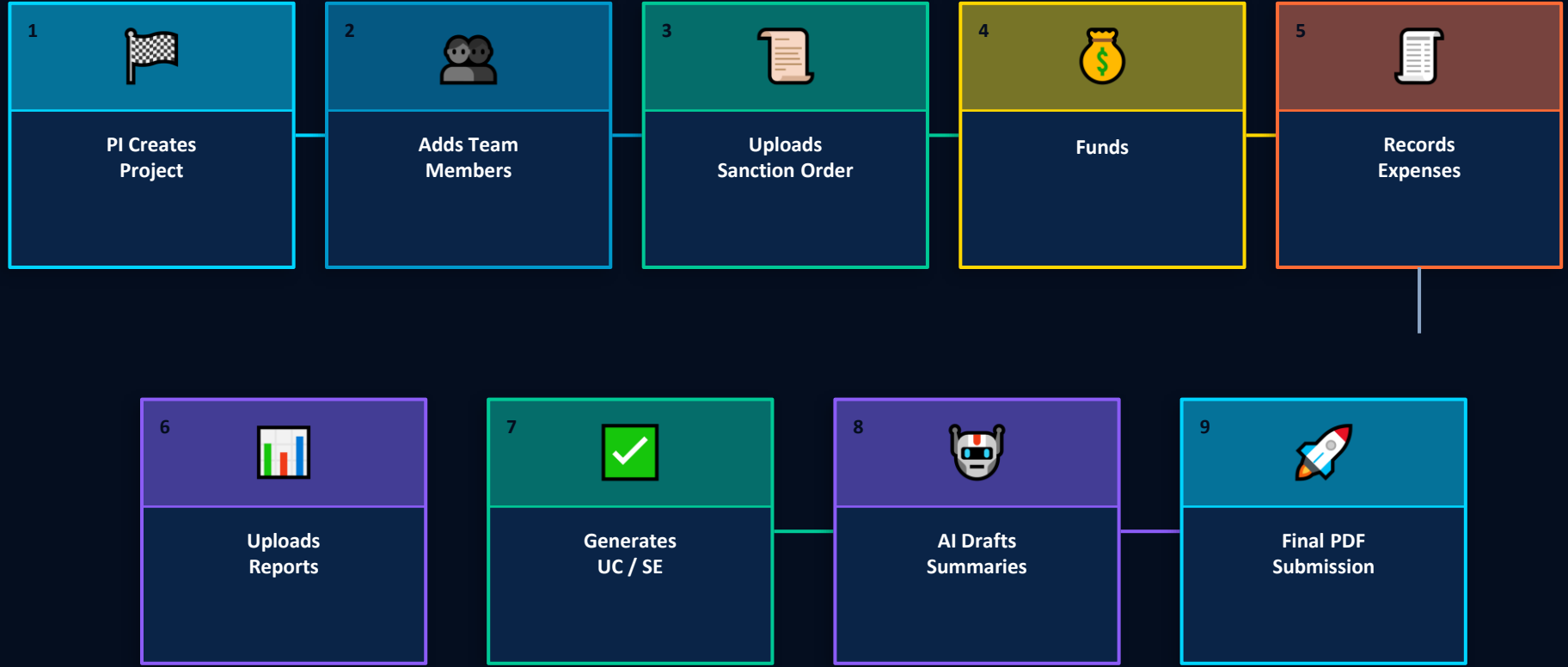
projects.end_date → coord: (120, 220)
Project end date stamp

AI INTEGRATION — Intelligent Research Assistant



💡 *AI assists the PI — it does not replace human judgment. All AI output is reviewed before official submission.*

COMPLETE END-TO-END WORKFLOW



THANK YOU